

Cloud Security (WAF) Set-up: Protecting Your Web Applications

Practical guide to selecting, deploying and operating Web Application Firewalls for cloud environments. Concise, visual, and focused on real-world configuration, optimisation and monitoring.

THEME: CLOUD WAF

COLOURS: #E7BF8AFF, #DFA858, #E6CC75



WAF

HTTP Request path

HTTP request path

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INSPECTION

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VS
BECONSED

RESPNSES

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Understanding Web Application Firewalls (WAFs)

WAFs inspect HTTP(S) traffic to block malicious requests before they reach your application. They operate at the application layer, enforce policies, and protect against common web attacks while preserving legitimate traffic.



Core function

Detect and block malicious payloads in requests (SQLi, XSS, LFI).



Modes

Blocking, monitoring (observe only), and learning/mode for rule tuning.



Cloud fit

Cloud-native WAFs integrate with CDNs and serverless apps; others are appliance or managed services.

Common Web Application Threats

Recognise the most frequent attack types WAFs defend against so you can map rules and detection.



SQL Injection

Pattern detection for suspicious SQL tokens and input validation failures.



Cross-Site Scripting (XSS)

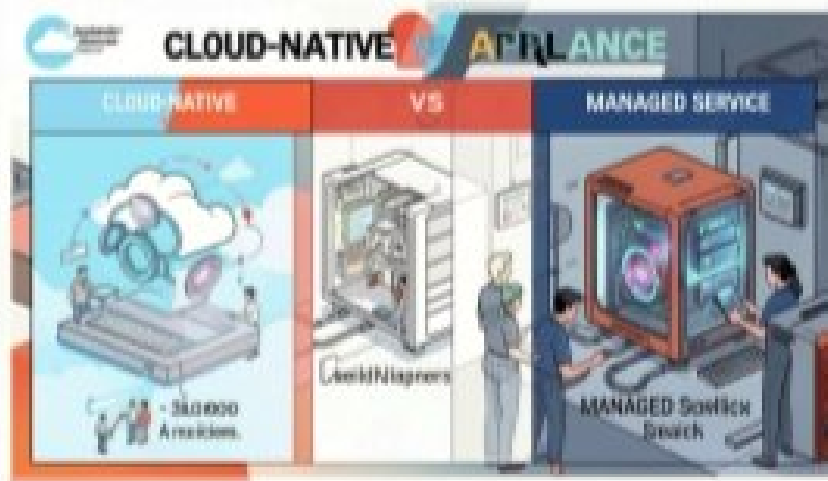
Sanitise outputs and filter script payloads in user inputs.



DDoS

Rate-limiting, challenge pages and integration with CDN/edge mitigations.

Choosing the Right WAF Solution

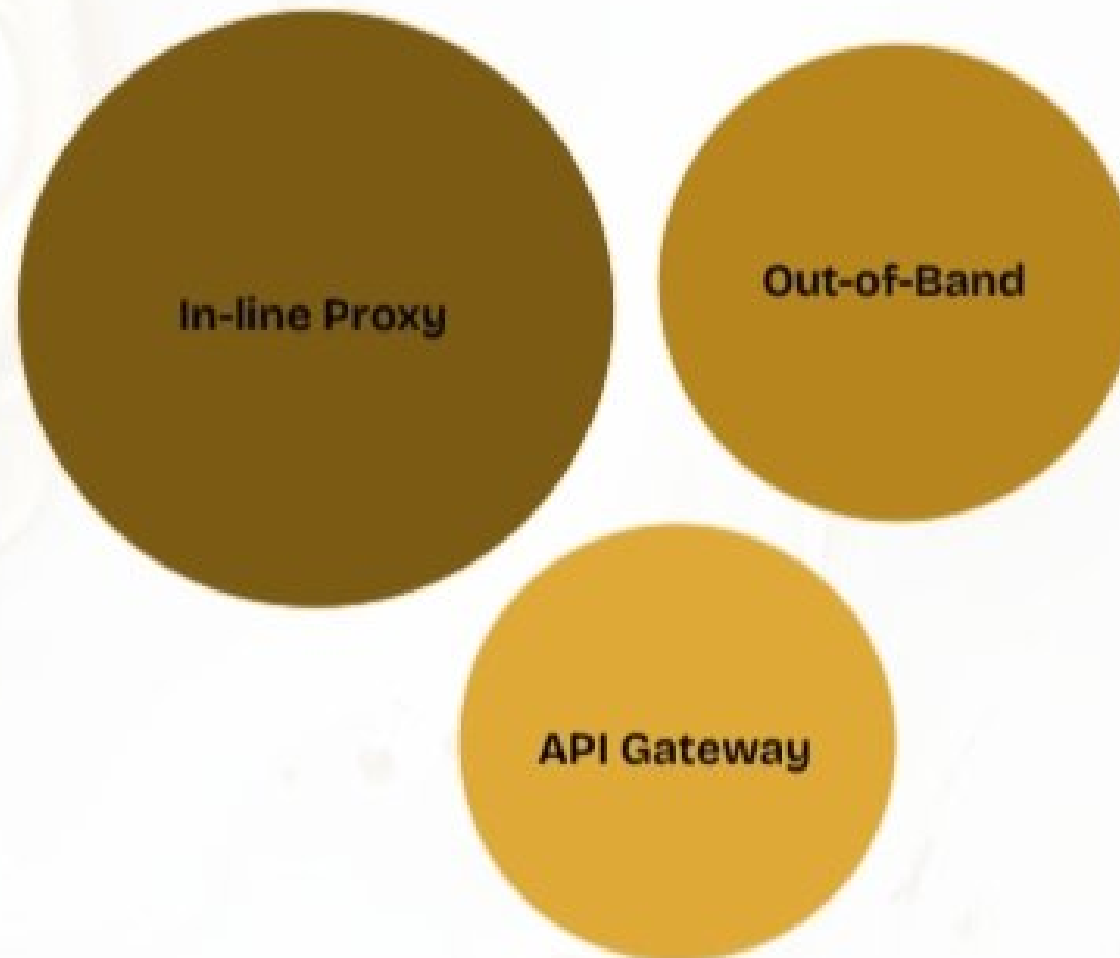


Decide based on scale, control and team capacity:

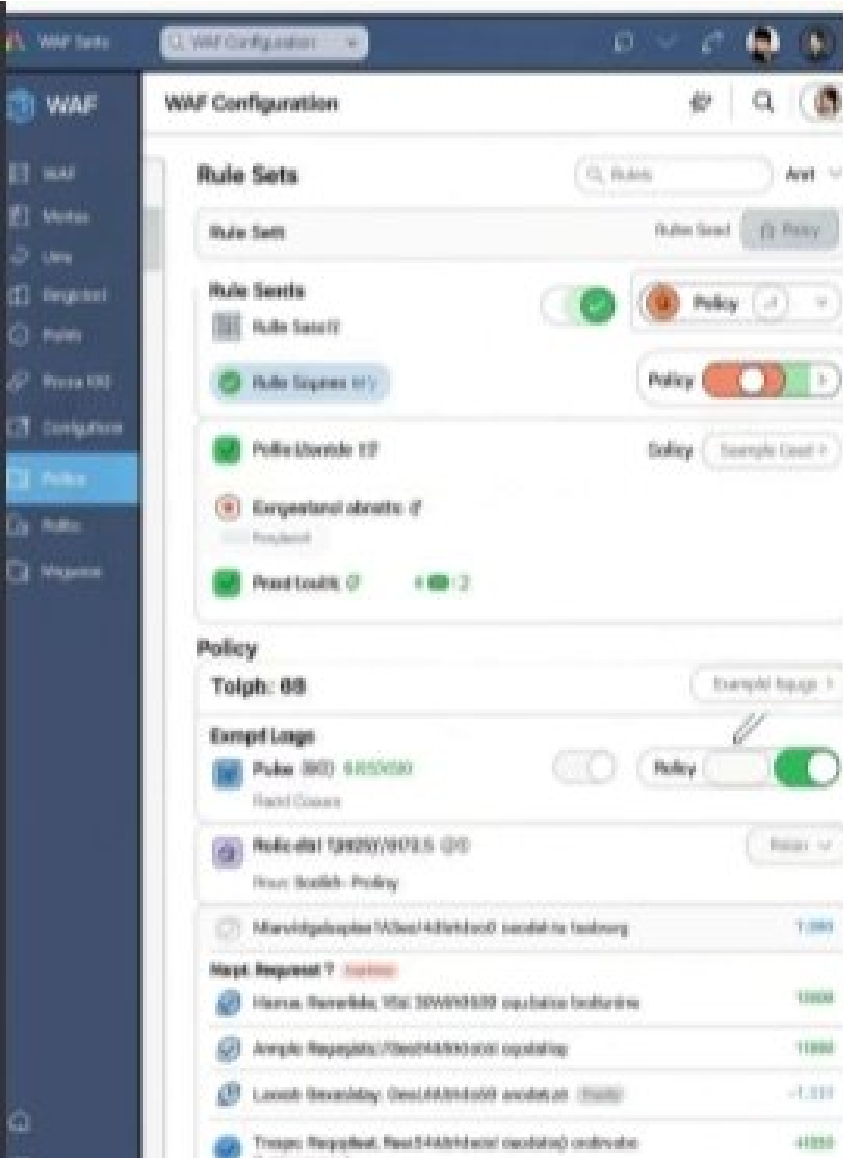
- Cloud-native: fast deployment, autoscaling, tight CDN/CDN integration.
- Appliance / virtual: full control, suitable for on-prem or private clouds.
- Managed service: vendor handles tuning and incidents — best for limited ops.

☐ Use hybrid models where traffic routing or compliance demands it.

WAF Deployment Architectures



Choose an architecture that fits latency tolerance and failure modes. In-line blocks immediately; out-of-band observes and reports; API gateways protect microservices and serverless endpoints.



Initial WAF Configuration

Start with a safe baseline and iterate:

Policy creation

Define application groups, enforce TLS, and create environment-specific policies.

Rule sets

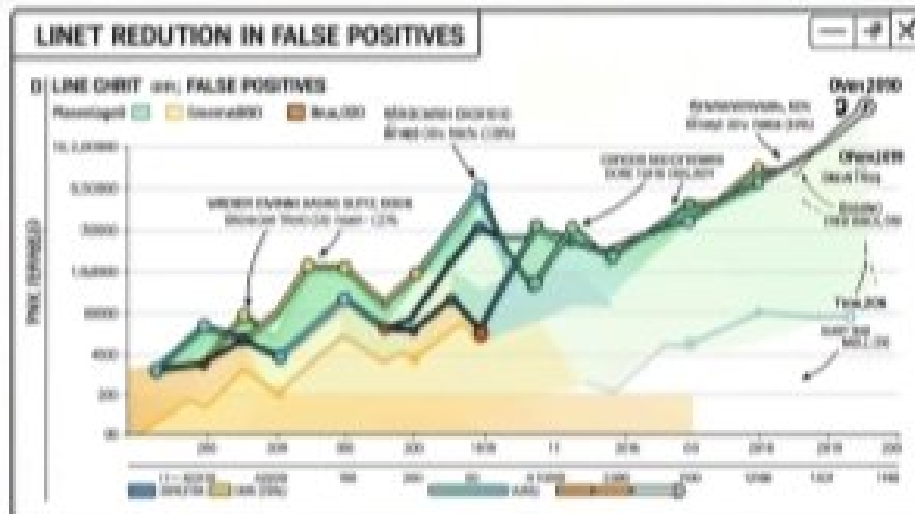
Enable vendor OWASP CRS, add application-specific custom rules for known endpoints.

Anomaly detection

Enable behavioural learning to reduce false positives during warm-up.

Tip: Put WAF in monitoring mode first for 7-14 days to gather baseline traffic patterns.

Fine-tuning and Optimisation

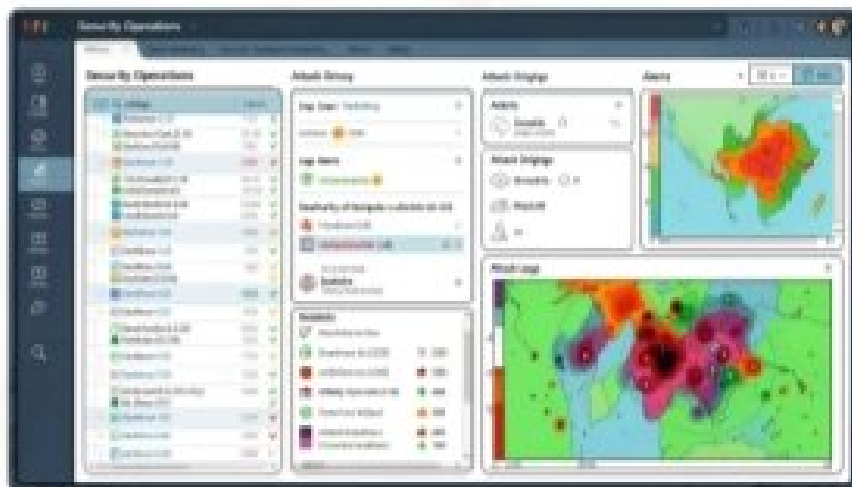


Optimise rules and performance:

- Review blocked request samples weekly; create precise allowlists or regex-based exclusions.
- Implement rate limits and bot profiles to separate benign spikes from attacks.
- Measure latency impact and enable edge caching to offset WAF processing time.

⚠️ Avoid over-permissive exceptions — they reduce protection.

Monitoring and Reporting

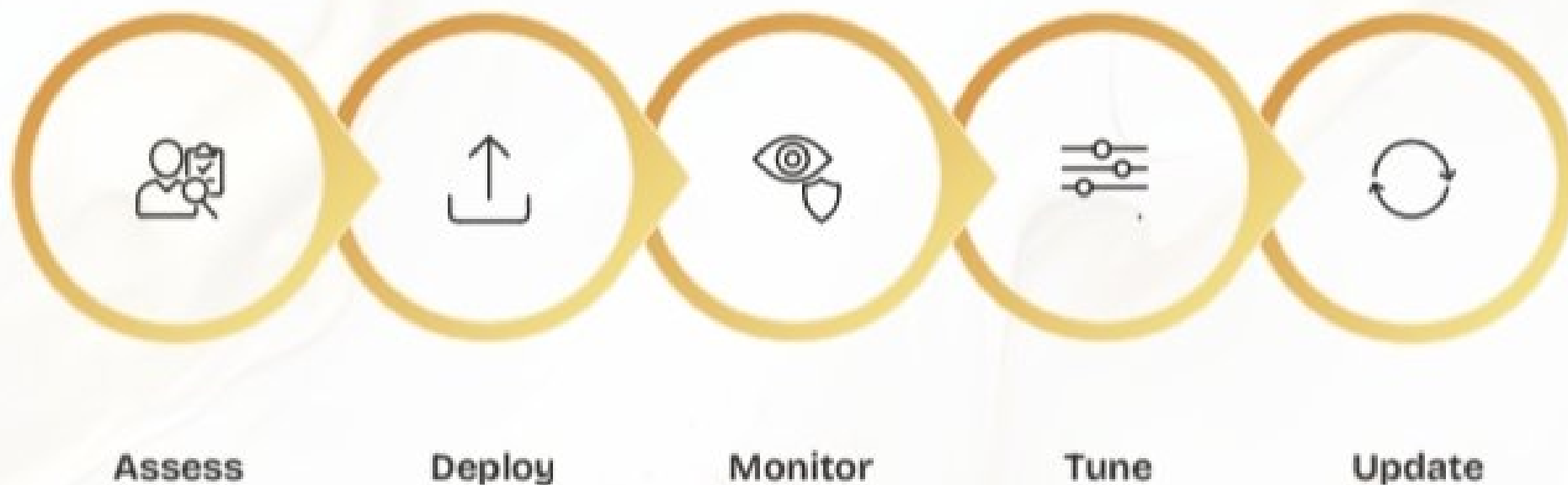


Critical telemetry and workflows:

- Centralised logging (SIEM) for request traces, alerts and forensic evidence.
- Real-time alerts for high-severity rule triggers and traffic anomalies.
- Scheduled reports: weekly incident summaries and monthly trend analysis.

☐ Integrate WAF logs with incident response playbooks to speed remediation.

Best Practices for WAF Management



Maintain a continuous cycle of assessment, tuning and response. Keep signatures and threat intelligence feeds up to date, test rules after changes, and run regular incident response drills.

Key Takeaways & Next Steps



Quick roadmap

Pick the deployment model, run WAF in monitoring mode, integrate logs, then transition to blocking once confident.



Operational readiness

Document policies, assign owners for rule reviews, and schedule quarterly tuning cycles.



Continuous improvement

Use metrics: false positive rate, mean time to detect/respond, attack volume trends to guide investments.

Ready to proceed? Build a pilot for a non-critical app, validate detection, then scale to production.

✔ Contact your security team to schedule a WAF pilot within 2–4 weeks.